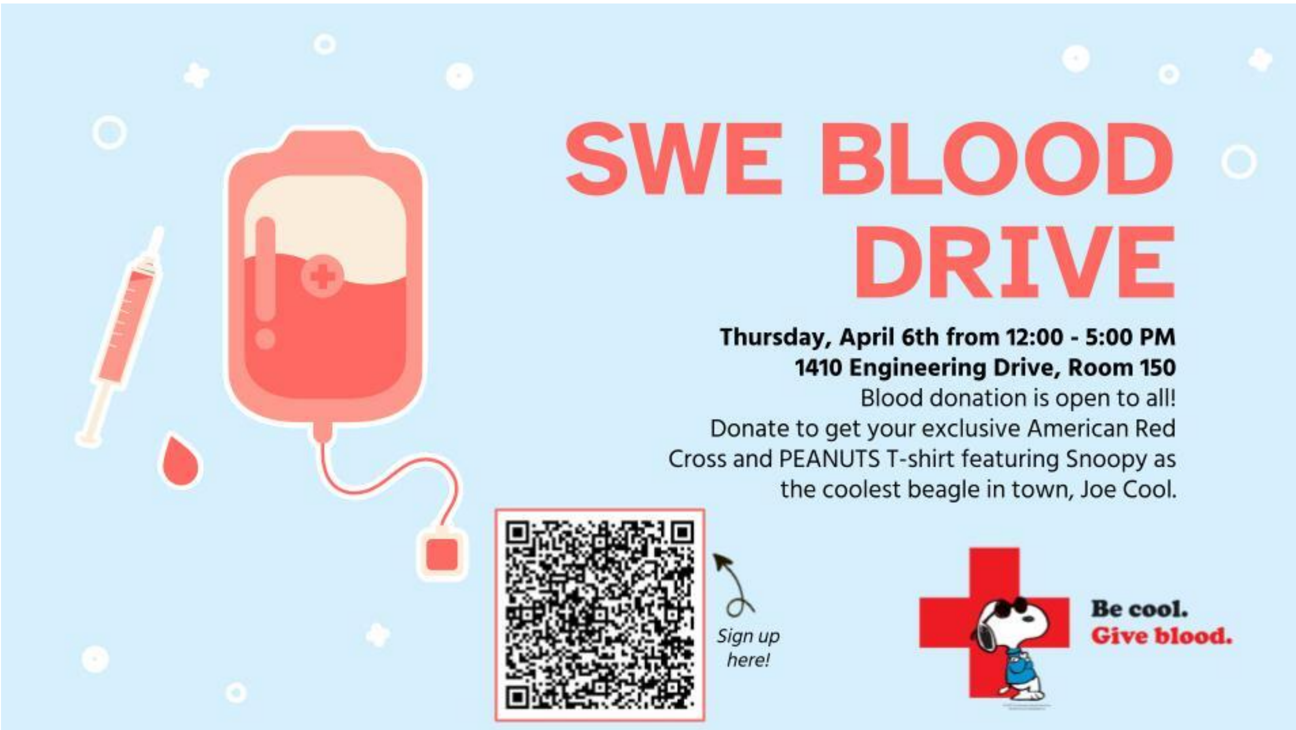
<u>Agenda</u>

1. Section 11.6
2. Section 11.7



Root Test

taking root of n^{th} power

see: n to the n^{th} power $\rightarrow$ root test

Suppose $\sum a_n$ is a series

1.) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} < 1$, then $\sum a_n$ absolutely converges.

2.) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} > 1$, then $\sum a_n$ diverges.

3.) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$, the test is inconclusive.

Example: Apply the root test to $\sum_{n=1}^{\infty} \left(\frac{3n^2-2}{n^2+3}\right)^{2n}$

$$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{3n^2-2}{n^2+3}\right)^{2n}\right|} = \lim_{n\to\infty} \sqrt[n]{\left(\left(\frac{3n^2-2}{n^2+3}\right)^2\right)^n}$$

$$= \lim_{n\to\infty} \left(\frac{3n^2-2}{n^2+3}\right)^2 = 3^2 = 9 > 1$$

$\sum_{n=1}^{\infty} \left(\frac{3n^2-2}{n^2+3}\right)^{2n}$ diverges by the root test

Activity: Apply the root test to $\sum_{n=2}^{\infty} \frac{(-1)^n}{(\ln n)^n}$

$$\lim_{n\to\infty} \sqrt[n]{\left|\frac{(-1)^n}{(\ln n)^n}\right|} = \lim_{n\to\infty} \sqrt[n]{\frac{1}{(\ln n)^n}} = \lim_{n\to\infty} \frac{1}{\ln n} = 0 < 1$$

$\sum_{n=2}^{\infty} \frac{(-1)^n}{(\ln n)^n}$ converges via the root test.

	Test	Convergence	Divergence	When to Use
1	Divergence	X	If $\lim_{n\to\infty} a_n \neq 0$, $\sum a_n$ diverges.	Always use it as a quick first check.
2	Geometric $\sum_{n=1}^{\infty} ar^{n-1}$	$\lvert r\rvert < 1$ converges to $\frac{a_1}{1-r}$	$\lvert r\rvert \geq 1$	When we can write a_n as ar^{n-1}.
	P-series	$p>1$	$p \leq 1$	When we can write a_n as $\frac{1}{n^p}$.
	(Direct) Comparison $0 \leq a_n \leq b_n$	If $\sum b_n$ converges, then $\sum a_n$ converges	If $\sum b_n$ diverges, then $\sum a_n$ diverges	When series are rational functions, similar to a p-series, similar to a geometric series.
	Limit Comparison $a_n, b_n \geq 0$, $\lim_{n\to\infty} \frac{a_n}{b_n} = c \neq 0$	If $\sum a_n$ converges, then $\sum b_n$ converges	If $\sum a_n$ diverges, then $\sum b_n$ diverges	Same as direct comparison ↑
		$\sum a_n$ and $\sum b_n$ do the same thing		
	Ratio	If $\lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert < 1$	If $\lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert > 1$	factorials
	Alternating $\sum (-1)^{n-1} a_n$	$\{a_n\}$ is positive, increasing, $\lim_{n\to\infty} a_n = 0$ ★ check for absolute convergence first	X	$\sum (-1)^{n-1} a_n$
	Root	If $\lim_{n\to\infty} \sqrt[n]{\lvert a_n\rvert} < 1$	If $\lim_{n\to\infty} \sqrt[n]{\lvert a_n\rvert} > 1$	$a_n = (b_n)^n$
	Integral $a_n = f(n)$ and is positive, decreasing, continuous	If $\int_1^{\infty} f(x)\,dx$ converges	If $\int_1^{\infty} f(x)\,dx$ diverges	When $\int_1^{\infty} f(x)\,dx$ is easy to integrate.

<u>Activity</u>: Determine whether each series converges (absolutely or conditionally) or diverges. Use any applicable test.

1.) $\sum_{K=1}^{\infty} \frac{K+4}{K\sqrt{3K+2}}$

comparison $\frac{K}{K^{3/2}}$ or $\frac{K}{K\sqrt{K}}$

2.) $\sum_{n=1}^{\infty} \frac{9n^3+5n^2}{n^{5/2}+4}$

divergence $(3 > 5/2)$

3.) $\sum_{n=1}^{\infty} \frac{3^{2n-1}}{n^2+2n}$

ratio

4.) $\sum_{n=1}^{\infty} \frac{n!}{(2n)!}$

ratio

5.) $\sum_{K=1}^{\infty} \left(\frac{-1}{K}\right)^K$

root

6.) $\sum_{n=1}^{\infty} n^3 e^{-n^4}$

integral

7.) $6+2+\frac{2}{3}+\frac{2}{9}+\frac{2}{27}+\cdots$

geometric

8.) $\frac{1}{3}-\frac{2}{4}+\frac{3}{5}-\frac{4}{6}+\cdots$

alternating